

Introduction to Cryptocurrency Markets

FINS5545: FINANCIAL MARKET DATA LITERACY

WHAT THIS LECTURE COVERS

1. **Why study cryptocurrency** — the opinions, the bubble question, and what makes it a distinctive market.
2. **Market structure** — thousands of coins and hundreds of exchanges.
3. **Institutional adoption** — futures, spot ETFs, and corporate treasuries.
4. **Market pricing** — exchange rates, quoting conventions, and computing returns.
5. **Advanced topics** — stablecoins, decentralised finance, and regulation.

WHY STUDY CRYPTOCURRENCY?

THE EXPERTS CANNOT AGREE — THE SCEPTICS

Few assets divide informed opinion as sharply as Bitcoin. The sceptics...

- **Nouriel Roubini** (economist who called the 2008 crisis) — “the biggest bubble in human history” and “an environmental disaster”.
- **Jamie Dimon** (CEO of JPMorgan) — “it is a fraud” and “worse than tulip bulbs”.
- **Jack Bogle** (founder of Vanguard, who created the index fund) — “avoid Bitcoin like the plague. There is nothing to support Bitcoin except the hope that you will sell it to someone for more than you paid for it.”

THE EXPERTS CANNOT AGREE — THE BELIEVERS

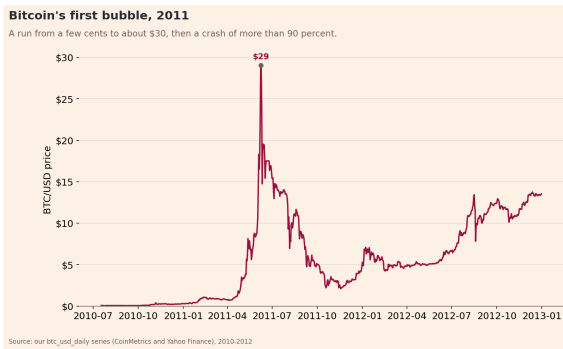
Now the believers — people just as accomplished, arguing the opposite.

- **Nassim Taleb** (author of *The Black Swan*) — “the beginning of something great: a currency without a government”. He later reversed his view and called it worthless, which tells you how unsettled even expert opinion is.
- **Eric Schmidt** (CEO of Google, 2001–2011) — “a remarkable cryptographic achievement ... with enormous value”.
- **Tyler Winklevoss** (co-founder of the Gemini exchange) — “a mathematical framework that is free of politics and human error”.

People at the very top of finance and technology disagree completely, and some change their minds. Form your own view from the evidence rather than from headlines.

IS IT A BUBBLE...?

Bitcoin's first mania. In 2011 the price ran from a few cents to about \$30, then collapsed more than 90 percent.



On its own, a textbook bubble — a steep climb to a peak, then a crash.

IS IT A BUBBLE?

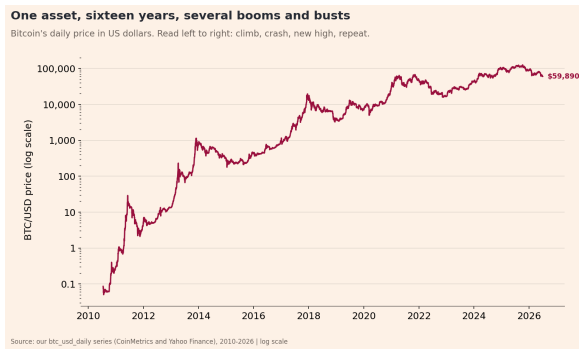
A **bubble** is a price that climbs far above any reasonable value, then collapses and does not come back. Several Bitcoin bull runs, looked at on their own, fit that description exactly.



Each run is a steep climb to a peak, then a crash of 75 to 85 percent. In isolation, every one of them looks like a classic bubble.

HISTORY SEEMS TO RHYME

Now every day of Bitcoin's price on one chart, from that first bubble to today.



The same shape — climb, crash, then a new high — repeats cycle after cycle.

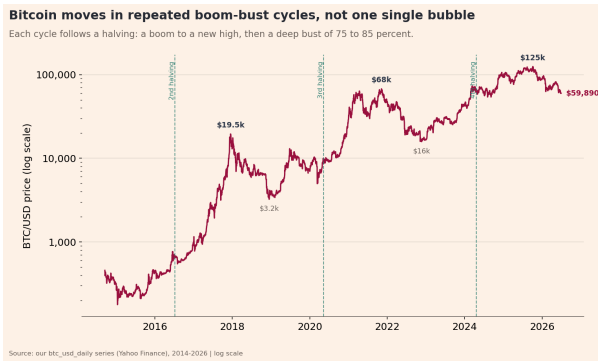
THE INCEPTION OF BUBBLES



Bubble within a bubble within a bubble...?

OR A REPEATING CYCLE?

A true bubble pops once and dies. Put the same runs on one log-scale chart and a different shape appears.

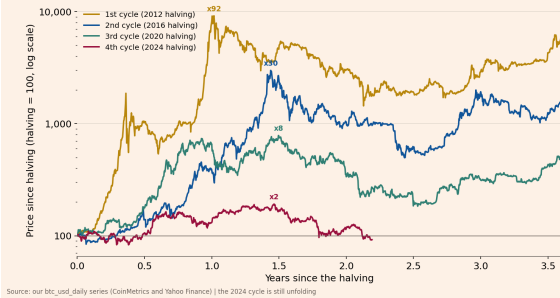


Four booms, four crashes of 75 to 85 percent, four recoveries to a new high. Each boom follows a **halving**, the roughly four-yearly halving of the reward paid to Bitcoin miners, which cuts the supply of new coins.

THE CYCLES RHYME

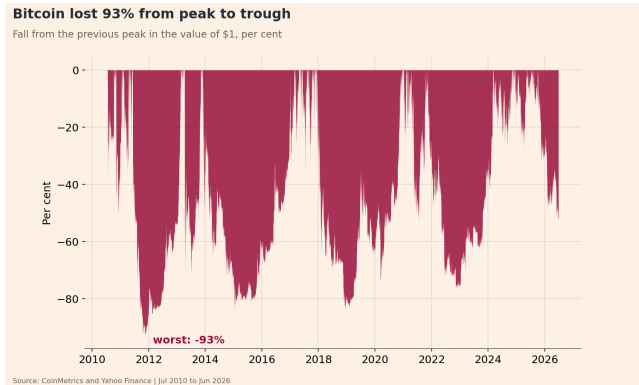
The cycles rhyme, but each boom is smaller than the last

Each cycle indexed to 100 at its halving. The timing repeats; the peak multiple shrinks every time.



Indexed to each halving, the timing repeats — a peak 12 to 18 months later — but each boom is smaller than the last, from about ninety-fold in the 2013 cycle to thirty, then eight, then two. The pattern is a cycle, not a one-time mania.

THE DRAWDOWNS ARE GETTING LESS BAD



The **drawdown** is the fall from the previous peak. The worst crash in each cycle has grown shallower — about **-93%** in 2011, into the eighties in the next two cycles, then about **-77%** in 2022 and roughly **-52%** so far in the 2025 to 2026 fall. As the market deepens, the busts hurt less, though they are still severe.

SO, BUBBLE OR NOT?

- A classic bubble — 17th-century tulips, or a dot-com firm that went bankrupt — pops once and never recovers.
- Bitcoin has fallen about 80 percent **four times** and reached a new all-time high in the next cycle **every time**, most recently to a record \$124,753 on 6 October 2025.
- That does not make it safe or correctly valued. The crashes are brutal, and the size of each boom is clearly shrinking.
- It does mean the simple story “it is just a bubble” does not fit twelve years of data. A volatile, adoption-driven asset moving on a four-year supply cycle fits better.

Treat Bitcoin as a high-volatility cyclical asset, not a single bubble waiting to pop. The real question is not “will it pop” but “where are we in the cycle”.

WHAT THE RESEARCH FINDS

- Statistical tests built to detect bubbles date-stamp *several* explosive run-ups in Bitcoin, not one terminal mania (Cheung et al., 2015; Corbet et al., 2018). Repeated bubbles are the statistical signature of a **cycle**.
- At the same time, Cheah and Fry (2015) find a large speculative component and argue Bitcoin's fundamental value is hard to justify. Both can be true at once — cyclical *and* speculative.
- One caveat. Some analysts now argue the four-year cycle is flattening as the asset matures and its volatility falls, so the next cycle may look milder than the last.

A DISTINCTIVE MARKET FOR STUDYING ASSET PRICING

Beyond the price, crypto is a useful laboratory because its market structure is unlike equities.

1. **Fragmented and granular.** Hundreds of exchanges, website brokers, and over-the-counter desks, with no single central market maker.
2. **Heterogeneous participants.** Miners, retail traders, large investors, and high-frequency firms all trade side by side.
3. **Continuous trading.** The market runs 24 hours a day, 7 days a week, so information arrives without the overnight gaps that equities have.
4. **Lightly regulated.** Historically exposed to manipulation, from wash trading to coordinated pump-and-dump schemes.

A HYBRID ASSET CLASS

Cryptocurrencies borrow features from several traditional asset classes at once, which is part of why they are hard to value.

1. Every coin has a **market capitalisation**, like a stock (price times coins in circulation).
2. Many coins are meant to work as a **means of payment**, like a currency.
3. Coins are quoted as a price of one currency in terms of another, exactly like **foreign exchange**.
4. Empirically, cryptocurrency returns have behaved more like foreign-exchange returns than like equity returns.

THE SIZE OF THE MARKET TODAY

- All cryptocurrencies together are worth about **\$2.1 trillion** in mid-2026, down from roughly \$2.4 trillion at the start of the year.
- **Bitcoin** alone is about **57 percent** of that. Its share of the total is called its **dominance**. Bitcoin and Ethereum together are about two-thirds of the crypto market.
- For scale, \$2.1 trillion is a large company but a small slice of global markets — roughly the size of one mega-cap technology stock, and a fraction of the US equity or bond market.

The whole crypto market is smaller than a single big-tech stock, and most of its value sits in just two coins.

STABLECOINS, THE DOLLAR OF THE CRYPTO MARKET

A **stablecoin** is a coin pegged to a fiat currency, almost always the US dollar. It is how most crypto trading is actually settled.

- **Tether (USDT)** is the largest, about **\$186 billion** in mid-2026. **USDC** is about \$74 billion. All stablecoins together are about \$310 billion.
- Most crypto-to-crypto trades are quoted against USDT, the way foreign exchange is quoted against the US dollar. USDT is now the most-traded cryptocurrency by volume.
- The long-running question is whether Tether's reserves fully back the coins it has issued. That transparency concern has followed it for years.

MOST TRADING VOLUME IS DERIVATIVES

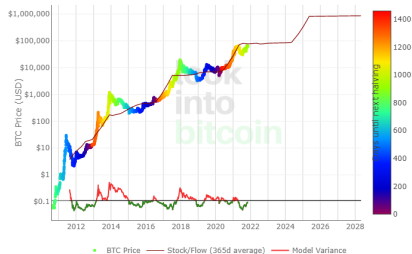
- Buying the coin itself is **spot** trading. Betting on its price with futures and perpetual swaps is **derivatives** trading.
- In mid-2026, derivatives are roughly **80 to 90 percent** of volume on the large centralised exchanges, several times the spot volume.
- Derivatives volume is concentrated in a few venues. Binance, OKX, Bybit, Bitget, and Gate handle most of it, with Binance alone about a third.

HOW DO YOU VALUE A COIN WITH NO CASH FLOWS?

- A stock has earnings and dividends, and a bond has coupons. Bitcoin has neither, so the standard valuation tools do not apply.
- That vacuum has been filled by eye-catching, chart-based “models” that promise to tell you what Bitcoin is worth and where its price is going.
- Two of them became famous — stock-to-flow and the rainbow chart. We look at both.
- Both are cautionary tales about confusing a good-looking chart with a real model.

STOCK-TO-FLOW, THE MODEL AND ITS PROMISE

The **stock-to-flow (S2F)** model (PlanB, 2019) borrows an idea from gold. Scarcity is the **stock** (coins already mined) divided by the **flow** (new coins each year). Every halving roughly doubles the ratio.



Regressing the log price on log stock-to-flow gives an eye-popping fit (about $R^2 = 0.95$) and a bold forecast — roughly \$100,000 average for the 2020–2024 cycle, and \$288,000 in a later version. The fit looks almost too good to be true — and it is.

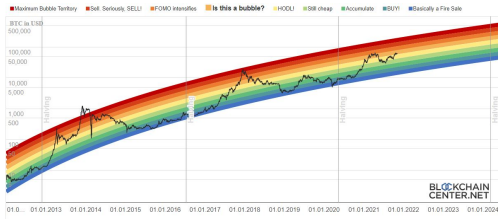
- Both series only trend upward over time, so each has a **unit root** — the fit is a **spurious regression**, a shared upward trend mistaken for a real link.

WHY STOCK-TO-FLOW FAILS

- **The forecast missed badly.** It called for a \$100k average and \$288k fair value. Bitcoin peaked near **\$69,000** in 2021, then crashed about 80 percent. It only reached six figures in the *next* cycle, still far below \$288k.
- **The good fit is an illusion.** Both the price and the scarcity ratio rise over time, so regressing one on the other gives a high R^2 even with no real link. This is the textbook spurious regression.
- **The model partly explains itself.** Market value is price times stock (the coins mined so far), and the scarcity ratio already contains that same stock, so the same quantity sits on both sides of the regression. Account for that overlap and the relationship largely breaks down.
- **It ignores demand.** Price is set by supply *and* demand. A supply-only model cannot value the asset.

A 0.95 R^2 sold a story. The model failed its first real out-of-sample test, then the goalposts moved.

THE RAINBOW CHART



The **rainbow chart** fits a curve through past prices and paints colour bands around it, from “fire sale” at the bottom to “maximum bubble” at the top.

- Its own creators call it a joke. The site that hosts it states it was “never a serious attempt to model or predict the price” and “there is no scientific basis”.
- The bands are eyeballed, the curve is re-fitted whenever the price breaks out, and the log scale shrinks 80 percent crashes into gentle dips.

A chart that looks quantitative — a regression, a log scale, neat bands — with no theory behind it. A clean case of overfitting and hindsight.

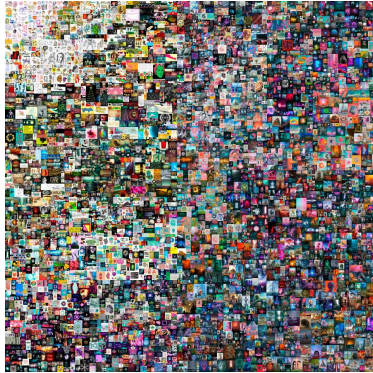
THE STRUCTURE OF THE MARKET

THOUSANDS OF COINS, BUT VALUE IN A FEW

The crypto market has thousands of assets.

- **2009** Bitcoin launches. **2015** Ethereum adds programmable **smart contracts**. **2017** the ICO boom floods the market with new tokens. **2020–21** decentralised finance and digital collectibles (NFTs) explode.
- Counting is messy. CoinGecko tracks about **16,000 to 17,000** coins and CoinMarketCap lists over 10,000. Millions more dead or scam tokens have come and gone.
- Yet the **top ten** coins are roughly **90 percent** of all crypto value.

“EVERYDAYS: THE FIRST 5000 DAYS” BY BEEPLE (\$69.3 MILLION)



Sold at a Christie's auction in March 2021 to a collector. This single digital image is a **non-fungible token (NFT)**, a unique token on a blockchain that records who owns a digital item. It was bought for **69.3 million US dollars**.

BORED APES



Justin Bieber's **Bored Ape Yacht Club** NFT, one of 10,000 cartoon apes used as profile pictures. At the 2021–22 peak the cheapest ape still cost hundreds of thousands of dollars. Their price rested on demand alone, like the coins with no cash flows, and many later lost most of their value.

HUNDREDS OF EXCHANGES, TIGHTLY CONCENTRATED

- About **1,500** exchanges are tracked, though only around 100 to 180 are actively ranked by real volume.
- There are two kinds. A **centralised exchange (CEX)** such as Binance, Coinbase, Kraken, or OKX. A **decentralised exchange (DEX)** such as Uniswap or PancakeSwap.
- Concentration is high. **Binance** alone is about **40 percent** of global spot volume, and the top five exchanges handle most of the rest.
- Exchanges relocate across borders to follow lighter rules, a practice called **regulatory arbitrage**.

CENTRALISED VERSUS DECENTRALISED EXCHANGES

Centralised (CEX)

- Traditional order book.
- Custodial — the exchange holds your coins.
- Higher liquidity, faster execution.
- Exposed to regulation and to hacks.

Decentralised (DEX)

- Automated market maker, no order book.
- Non-custodial — you keep your own coins.
- Runs on blockchain smart contracts.
- Lower liquidity, more slippage on large trades.

THE PATH TO LEGITIMACY

For its first decade, crypto sat outside mainstream finance.

- **2009–2016** Bitcoin was dismissed as money for criminals. There was no institutional infrastructure, deep regulatory uncertainty, and volatility that scared institutions away.
- Turning points... the first Bitcoin ATMs in 2013, regulated exchanges such as Coinbase in 2014, Japan recognising Bitcoin as a legal means of payment in 2016, and the major derivatives venues planning futures in 2017.

CHICAGO MERCANTILE EXCHANGE (CME) LAUNCHES BITCOIN FUTURES

- In December 2017, the CBOE and then the **CME** launched regulated Bitcoin **futures**. The CBOE later withdrew, and the CME became the main institutional venue.
- The CME added Bitcoin **options** in January 2020, then Ethereum futures and options in 2021 and 2022.
- By 2026, CME crypto futures volume was up about **40 percent** year on year and trading nearly around the clock. Regulated derivatives are now a core part of the market.

THE SPOT-ETF BREAKTHROUGH

- From 2013 to 2021, the SEC rejected every application for a **spot Bitcoin ETF** (a fund holding actual Bitcoin).
- On **10 January 2024** the SEC approved the first US spot Bitcoin ETFs — eleven funds at once, including BlackRock's **IBIT**.
- This let any investor hold Bitcoin through an ordinary brokerage account, with no wallets, keys, or exchanges to manage.
- Spot **Ethereum** ETFs followed in July 2024.

- The US spot Bitcoin ETFs hold about **1.25 million BTC**, roughly 6 percent of all Bitcoin, worth about \$75 to \$82 billion in mid-2026.
- BlackRock's **IBIT** is by far the largest, holding around **750,000 BTC** on its own.
- 2026 was their first bad year. As Bitcoin fell from \$125k toward \$60k, the ETFs saw sustained **net outflows** of several billion dollars, the first since launch. Ethereum ETFs hold about \$10 billion.

COMPANIES THAT HOLD BITCOIN

- **Strategy** (formerly MicroStrategy) holds about **847,000 BTC** in mid-2026, worth tens of billions of dollars, funded by issuing debt and shares. It holds more than every other public company combined.
- A wave of imitators, the **Bitcoin treasury companies**, copied the model worldwide. Tesla and others hold smaller positions.
- In 2025 the US government joined in, creating a **Strategic Bitcoin Reserve** by executive order.

A treasury company's shares become a leveraged bet on Bitcoin. When Bitcoin fell in 2026, Strategy made its first sale since 2022.

HOW CRYPTO PRICES ARE QUOTED

EXCHANGE RATES, BASE AND QUOTE

An **exchange rate** is the price of one currency in terms of another. It is quoted as units of the **quote** currency per one unit of the **base** currency.

$\underbrace{\text{BTC}}_{\text{base}} / \underbrace{\text{USD}}_{\text{quote}}$ “how many US dollars for one Bitcoin”

- The **base** is the currency you buy, the good.
- The **quote** is what you pay for it, the price.

QUOTING CONVENTIONS

- In **fiat-to-crypto** markets, the fiat currency is always the quote. The most liquid fiat pair moves between the US dollar and the Japanese yen.
- In **crypto-to-crypto** markets, the most common quotes are Bitcoin and **USDT**.
- USDT is now the most-traded quote of all, playing the role that the US dollar plays in foreign exchange.

COMPUTING A RETURN

The **rate of return** measures the gain or loss on an investment over a period. Using BTC/USD prices,

$$R_t = \left(\frac{P_t}{P_{t-1}} - 1 \right) \times 100.$$

This is the same formula you used for stocks. The only difference is that crypto trades every calendar day, so there are no weekend gaps in the price series.

WORKED EXAMPLE, BTC/USD

Suppose Bitcoin trades at \$20,000 on Day 1 and \$22,000 on Day 2.

$$R = \left(\frac{22,000}{20,000} - 1 \right) \times 100 = \mathbf{10\%}.$$

This is exactly the daily-return calculation you ran on real Bitcoin prices in the practise exercise.

ADVANCED TOPICS

WHAT MAKES CRYPTO MICROSTRUCTURE DIFFERENT

Versus traditional markets

- Trades 24/7, not business hours.
- Algorithmic makers, not designated specialists.
- Hundreds of venues, not a few.

Crypto-specific effects

- Cross-exchange arbitrage on price gaps.
- Flash crashes.
- Wash trading inflates volumes.
- Coordinated pump-and-dump schemes.

STABLECOINS AND THE NEW RULES

- Three kinds of stablecoin. **Fiat-backed** (USDT, USDC, held in dollars and Treasuries), **crypto-backed** (DAI), and **algorithmic** (Terra/UST, which collapsed to zero in 2022 and wiped out tens of billions of dollars).
- Most crypto trading settles in stablecoins. They are the working dollar of the market.
- 2025 brought additional regulation. The US **GENIUS Act**, signed in July 2025, sets the first federal rules for dollar stablecoins — full reserves, regular audits, and a ban on algorithmic stablecoins. Its rules take effect by early 2027.

THE REGULATORY MAP IN 2026

- **United States.** A sharp pro-crypto turn in 2025 — the GENIUS Act, a Strategic Bitcoin Reserve, a ban on a government digital currency, and a regulator that dropped many enforcement cases.
- **European Union.** The **MiCA** framework is now fully in force. From mid-2026, a firm must be licensed to serve EU clients.
- **Asia.** A patchwork, from full acceptance to outright bans.
- **Australia.** No single crypto law — existing rules apply. The corporate regulator treats many tokens and platforms as financial products that need a licence, exchanges must register for anti-money-laundering checks, and the tax office taxes crypto as property under capital-gains tax.

CONCLUDING REMARKS

WHAT TO TAKE AWAY

1. Take what mainstream media says about crypto with a pinch of salt. The experts themselves disagree, and some change their minds.
2. If you hold a view on crypto, can you back it with careful research and data rather than a headline?
3. Bitcoin looks less like a one-off bubble and more like a volatile asset on a roughly four-year cycle, although each boom has been smaller than the last.
4. Institutional adoption — futures, spot ETFs, corporate treasuries, and real regulation — has wired crypto into mainstream finance.

SOURCES I

Market size, coins, and exchanges (mid-2026)

- CoinGecko — total market cap, coins tracked, exchange count
- CoinMarketCap — historical market-cap snapshot, June 2026
- CoinGecko Research — top-ten concentration
- Datawallet — exchange volumes and Binance market share

Stablecoins, decentralised finance, and derivatives

- CoinMarketCap — USDT and USDC market caps, June 2026
- DeFiLlama — stablecoin supply and total value locked

Institutional adoption and regulation

- Bitbo — BlackRock IBIT Bitcoin holdings
- Strategy — corporate Bitcoin holdings
- White House — the GENIUS Act stablecoin law (July 2025)
- ESMA — the EU MiCA framework

Cycles and the critique of valuation models

- Fidelity — the four-year Bitcoin cycle and its drawdowns
- CoinGecko — the stock-to-flow model and its critics

- [Bitcoin Magazine](#) — why stock-to-flow is not useful
- [BlockchainCenter](#) — the rainbow chart and its disclaimer

Bitcoin price figures are computed from our own `btc_usd_daily` series (Yahoo Finance, 2014–2026).
Market figures are mid-2026 snapshots and move daily.

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- Adrian Cheung, Eduardo Roca, and Jen-Je Su. Crypto-currency bubbles: An application of the Phillips–Shi–Yu methodology on Mt. Gox Bitcoin prices. *Applied Economics*, 47(23):2348–2358, 2015. doi: 10.1080/00036846.2015.1005827.
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